Taseen Iqtider

Jose Avila

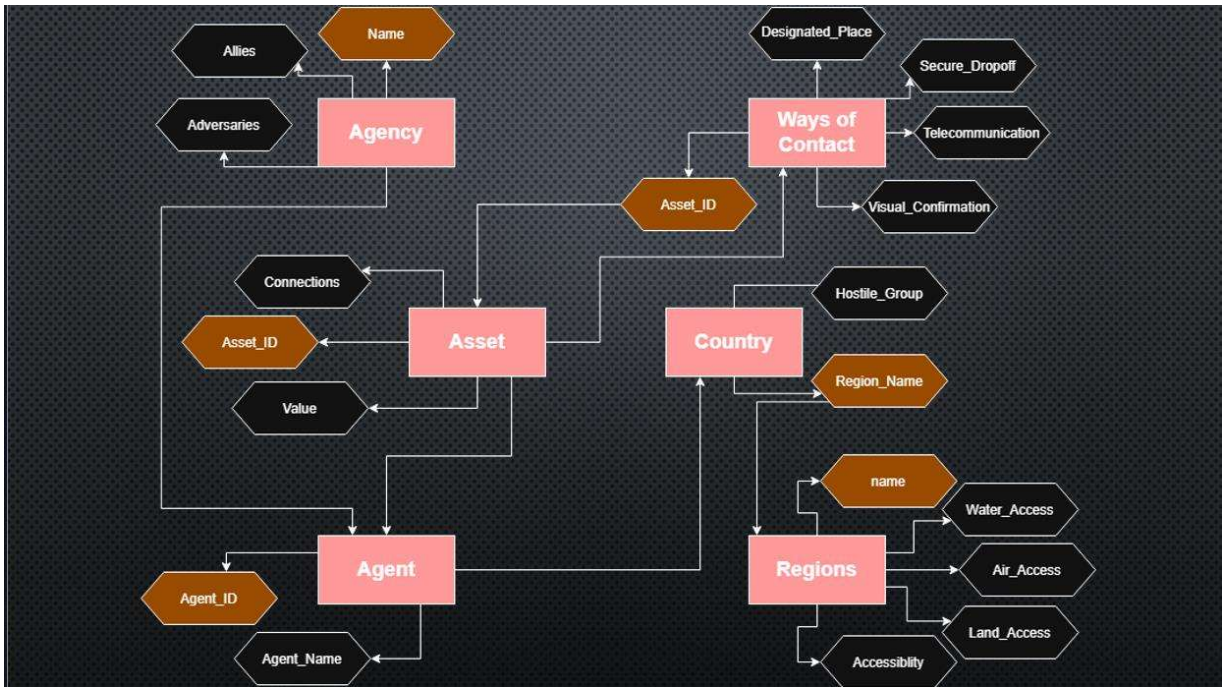
Austin Taylor

Milestone 2: Database Design and Milestone 3: Implementation

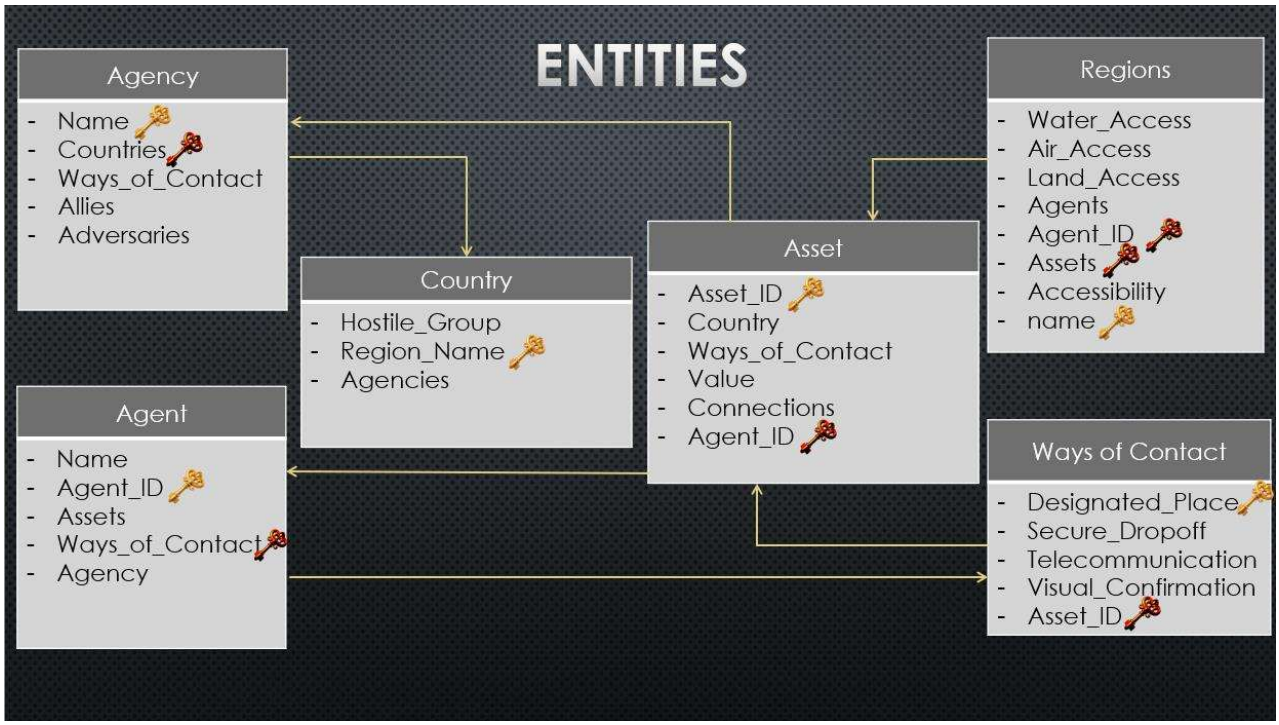
Requirements Analysis:

- Strong Encryption
- Secure information processing
- Understanding of lead and lag functions within MySQL
- Development of scrambling queries

ER Diagram:



Relational Schema:



Data Dictionary:

Region

Contains geographical areas and their overall access characteristics and accessibility status.

Column Name	Data Type	Key Status	Description
RegionName	VARCHAR	Primary Key	The name of the geographic region (e.g., North America, Western Europe).
Water_Access	VARCHAR (Enum/Boolean)	-	Indicates whether the region has access via water ('Yes'/'No').
Air_access	VARCHAR (Enum/Boolean)	-	Indicates whether the region has access via air ('Yes'/'No').
Land_access	VARCHAR (Enum/Boolean)	-	Indicates whether the region has access via land ('Yes'/'No').
Accessibility	VARCHAR	-	The general accessibility status of the region (e.g., Open, Restricted, Moderate).

Country

Stores individual countries, their associated region, and primary intelligence agencies.

Column Name	Data Type	Key Status	Description
CountryName	VARCHAR	Primary Key	The name of the country (e.g., USA, Canada).
CountryRegion	VARCHAR	Foreign Key	References Region(RegionName). The geographic region where the country is located.
CountryAgencies	VARCHAR	-	Primary intelligence agency associated with the country (can be NULL).

Ways of Contact

Details communication and coordination methods, including drop-offs, telecommunications, and visual cues used by assets and agents.

Column Name	Data Type	Key Status	Description
DesignatedPlace	VARCHAR	Primary Key	The specific location name or safe house used for contact/meetings.
Secure_Droppoff	VARCHAR	-	Physical dead drop method or location used for items (can be NULL).
Telecommunication	VARCHAR	-	Electronic communication method used, such as encrypted emails or burner phones (can be NULL).
Visual_Confirmation	VARCHAR	-	Physical sign or item used for visual identification during meetings (can be NULL).

Agency

Lists intelligence agencies, their home countries, standard contact locations, and diplomatic relations (allies and adversaries).

Column Name	Data Type	Key Status	Description
AgencyName	VARCHAR	Primary Key	Acronym or name of the intelligence agency (e.g., CIA, MI6).
AgencyCountries	VARCHAR	Foreign Key	References country(CountryName). The base country of the agency.

Column Name	Data Type	Key Status	Description
Contact_Place	VARCHAR	Foreign Key	References Ways_of_Contact(DesignatedPlace). Primary facility or safe house used by the agency.
Allies	VARCHAR	-	Comma-delimited list of allied intelligence agencies.
Adversaries	VARCHAR	-	Comma-delimited list of adversarial intelligence agencies.

Asset

Stores confidential informants, contacts, and assets operating within various countries, including their value and cover.

Column Name	Data Type	Key Status	Description
AssetID	INT	Primary Key	Unique numeric identifier for the asset.
AssetName	VARCHAR	-	Codename or first name of the asset.
AssetCountry	VARCHAR	Foreign Key	References country(CountryName). The home country of the asset.
AssetWays_of_Contact	VARCHAR	Foreign Key	References Ways_of_Contact(DesignatedPlace). The contact location assigned to the asset.
AssetValue	VARCHAR	-	Strategic intelligence value level of the asset (e.g., High, Medium, Low).
AssetConnection	VARCHAR	-	The asset's professional cover or access role (e.g., Government contact, Journalist).

Agent

Details field operatives working for specific intelligence agencies, including their assigned assets and operational contact points.

Column Name	Data Type	Key Status	Description
AgentID	INT	Primary Key	Unique numeric identifier for the field agent.
AgentName	VARCHAR	-	Full name of the intelligence agent.
AgentAssets	INT	Foreign Key	References Asset(AssetID). The primary asset managed by this agent.

Column Name	Data Type	Key Status	Description
AgentWays_of_Contact	VARCHAR	**Foreign Key**	References Ways_of_Contact(DesignatedPlace). The safe house or meeting spot utilized by the agent.
AgentAgency	VARCHAR	**Foreign Key**	References Agency(AgencyName). The intelligence agency the agent is employed by.

Normalization through 3NF: Yes

Sample Data:

INSERT VALUES:

```
use global_asset;


#region
INSERT INTO Region (Water_Access, Air_access, Land_access, Accessibility, RegionName)
VALUES
('Yes', 'Yes', 'Yes', 'Open',       'North America'),
('Yes', 'Yes', 'Yes', 'Open',       'Europe'),
('Yes', 'Yes', 'Yes', 'Restricted', 'Asia'),
('No',  'Yes', 'Yes', 'Restricted', 'Middle East'),
('Yes', 'Yes', 'Yes', 'Moderate',   'Africa');


#country
INSERT INTO country (CountryName, CountryRegion, CountryAgencies) VALUES
('USA',     'North America', 'CIA'),
('UK',      'Europe',        'MI6'),
('Germany', 'Europe',        NULL),
('Russia',  'Europe',        'FSB'),
('China',   'Asia',          'MSS'),
('Iran',    'Middle East',   NULL),
('France',  'Europe',        'DGSE'),
('Israel',  'Middle East',   'Mossad'),
('Egypt',   'Africa',        NULL),
('Japan',   'Asia',          NULL);


#ways_of_contact
```

```
INSERT INTO Ways_of_Contact (DesignatedPlace, Secure_Droppoff, Telecommunication,
Visual_Confirmation) VALUES
('Park Bench',    'Hollow tree',      'Burner phone',        'Red scarf'),
('Safe House',    'Locked cabinet',   'Encrypted email',     'Passphrase'),
('Cafe',          'Under table',      'Encrypted messaging', 'Newspaper'),
('Train Station', 'Locker 42',        'Burner phone',        'Blue umbrella'),
('Embassy Annex', 'Diplomatic pouch', 'Secure line',         'Badge check'),
('Hotel Lobby',   'Room safe',        'Secure line',         'Folded map'),
('Library',       'Book return slot', 'Encrypted messaging', 'Yellow folder');


#agency
INSERT INTO Agency (AgencyName, AgencyCountries, Contact_Place, Allies, Adversaries)
VALUES
('CIA',    'USA',    'Safe House',    'MI6', 'FSB'),
('MI6',    'UK',     'Embassy Annex', 'CIA', 'FSB'),
('FSB',    'Russia', 'Park Bench',    'MSS', 'CIA'),
('MSS',    'China',  'Cafe',          'FSB', 'CIA'),
('DGSE',   'France', 'Hotel Lobby',   'MI6', 'FSB'),
('Mossad', 'Israel', 'Library',       'CIA', 'MSS');


#asset
INSERT INTO Asset (AssetID, AssetName, AssetCountry, AssetWays_of_Contact, AssetValue,
AssetConnection) VALUES
(1,  'Victor', 'Russia',  'Park Bench',    'High',   'Government contact'),
(2,  'Ellen',  'Germany', 'Cafe',          'Medium', 'Embassy staff'),
(3,  'Faith',  'Iran',    'Train Station', 'High',   'Nuclear technician'),
(4,  'May',    'China',   'Cafe',          'High',   'Telecom engineer'),
(5,  'Daniel', 'USA',     'Park Bench',    'Medium', 'Defense contractor'),
(6,  'Claire', 'UK',      'Train Station', 'Low',    'Civil servant'),
(7,  'Kevin',  'Egypt',   'Hotel Lobby',   'High',   'Military officer'),
(8,  'Harry',  'Japan',   'Library',       'Low',    'Business executive'),
(9,  'Sam',    'Russia',  'Cafe',          'Medium', 'Bank clerk'),
(10, 'Lisa',   'Iran',    'Library',       'High',   'University professor');

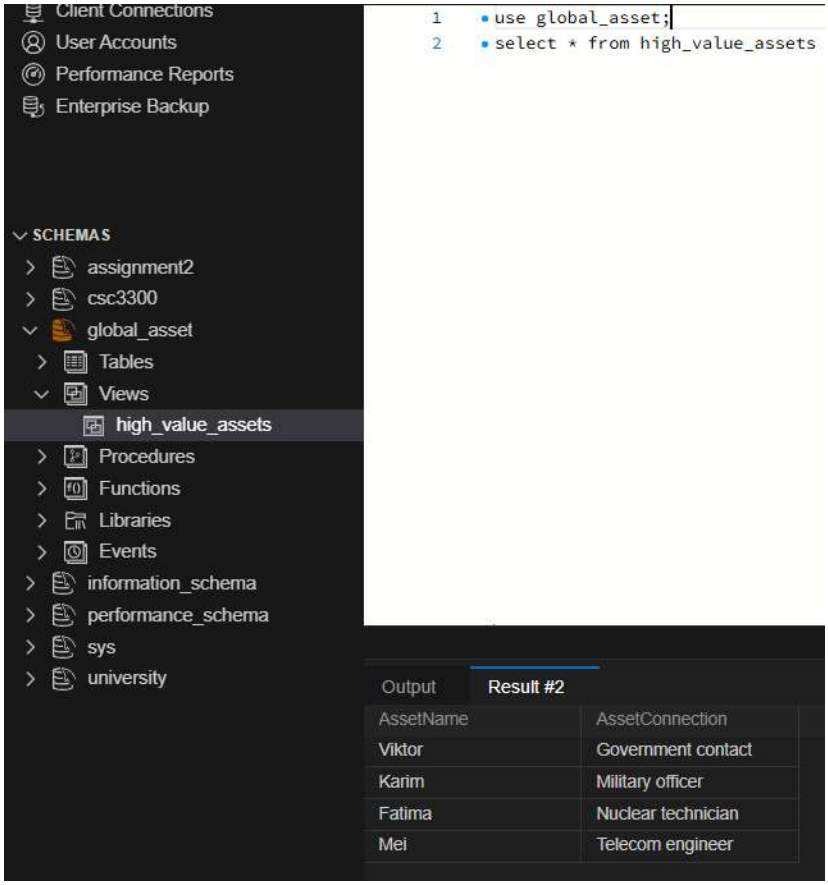

#agent
INSERT INTO Agent (AgentID, AgentName, AgentAssets, AgentWays_of_Contact,
AgentAgency) VALUES
(1,  'John Smith',      1,  'Safe House',     'CIA'),
(2,  'Mary Allen',      3,  'Train Station',  'CIA'),
(3,  'James Whitby',    2,  'Embassy Annex',  'MI6'),
(4,  'Olivia Hart',     4,  'Cafe',           'MI6'),
(5,  'Evan Peters',     5,  'Park Bench',     'FSB'),
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```
(6,  'Will Chambers',  6,  'Cafe',          'MSS'),
(7,  'Chris Davis',    7,  'Hotel Lobby',   'DGSE'),
(8,  'Rachel Cohen',   10, 'Library',       'Mossad'),
(9,  'Tom Becker',     9,  'Cafe',          'CIA'),
(10, 'Amy Tanner',     8,  'Train Station', 'MI6');
```

10 Meaningful queries that include aggregate functions, joins and subqueries:

1. Asset map: every asset with its country, region and how hard that region is to get into. (JOIN)
2. Handler briefing sheet: each agent, their agency, the asset they run, where they meet and the recognition signal. (Multi-table JOIN)
3. Compromised agent: if an agent is exposed, which asset and meeting place are now at risk. (JOIN, WHERE)
4. Burned meeting place: if a meeting place is discovered, which assets and handlers used it. (JOIN, WHERE)
5. Country crisis: when trouble breaks out in a country, which assets are there and who to contact. (JOIN, WHERE)
6. High-value watchlist: the most valuable assets and the agents responsible for them. (JOIN, WHERE)
7. Coverage gaps: countries in the database where you have no assets at all. (LEFT JOIN, IS NULL)
8. Regional strength: how many assets each region has, and how many are high value. (JOIN, COUNT, SUM, GROUP BY)
9. Agency workload: how many agents each agency has in the field. (COUNT, GROUP BY)
10. Agency value breakdown: how many High, Medium and Low value assets each agency runs. (JOIN, CASE, SUM, GROUP BY)
11. Overused meeting places: places used by more than one asset, which raises the risk of exposure. (COUNT, GROUP BY, HAVING)
12. No local agency: assets working in countries with no intelligence agency of their own. (Subquery with IN)
13. Riskiest assignments: agents whose asset is in a region with restricted access. (Nested subqueries)
14. Behind enemy lines: agencies running an asset inside an enemy's home country. (Correlated subquery with EXISTS)
15. Possible teamwork: pairs of agents from the same or allied agencies whose assets are in the same region. (SELF JOIN)

View:



Stored procedure/function and trigger: Not supported

Citations:

"Is this Normalization through 3NF?" *Gemini*, version 3.8, Google, October 2026

"What is an example of a data dictionary?" *Gemini*, version 3.8, Google, October 2026

"Data dictionary template" *Gemini*, version 3.8, Google, October 2026